

Business Problem Statement

Project Title: Customer Behavior Analysis Using Python and Power BI

Business Problem Statement

A retail company wants to understand customer purchasing behavior in order to:

- Increase overall revenue
- Improve customer retention
- Identify high-performing products and categories
- Optimize discount and marketing strategies
- Understand customer segments and subscription behavior
- Analyze seasonal and demographic purchasing trends

The company currently has raw customer transaction data but lacks meaningful insights that can support business decision-making.

The goal of this project is to clean, analyze, and visualize customer data to uncover patterns, trends, and actionable business insights.

Objectives of the Project

The main objectives of this Customer Behavior Analysis project are:

1. Analyze customer purchasing patterns and spending behavior.
2. Identify the product categories generating the highest revenue.
3. Evaluate the effectiveness of discounts on purchase value.
4. Compare spending behavior between male and female customers.
5. Identify customers who used discounts but still spent above-average amounts.
6. Analyze product ratings to determine top and bottom performing products.
7. Compare purchase values between Standard and Express shipping methods.
8. Understand the impact of customer subscriptions on revenue generation.
9. Identify products with the highest discount usage percentage.
10. Segment customers into New, Returning, and Loyal categories.
11. Analyze whether repeat buyers are more likely to subscribe.
12. Identify top-selling products within each category.
13. Study revenue contribution across different age groups.

Expected Outcomes

After completing this project, the retail company will be able to:

- Understand customer behavior more effectively
- Identify profitable customer segments
- Improve customer retention strategies
- Optimize discount campaigns
- Increase sales through targeted marketing
- Improve inventory and product planning
- Make data-driven business decisions

Key Business Questions Addressed

This project aims to answer the following business questions:

- Which category generates the highest revenue?
- Are discounts increasing purchase value?
- What is the revenue generated by male and female customers?
- Which customers spent above average despite using discounts?
- What are the top and bottom 5 products by review rating?
- How does shipping type affect average purchase value?
- Do subscribed customers spend more?
- Which products have the highest discount usage?
- What is the distribution of customer segments?
- Are repeat buyers more likely to subscribe?
- What are the top-selling products in each category?
- Which age group contributes the highest revenue?

Conclusion

This project demonstrates how data analytics can help retail businesses gain valuable insights into customer behavior. By combining Python-based data analysis with Power BI visualizations, the project enables businesses to improve decision-making, increase profitability, and build stronger customer relationships through data-driven strategies.